

OpenMRS

Sustainable Open Source Healthcare for the Developing World

This article offers a glimpse into OpenMRS, its origins, ideology and its recipe for success. It should interest open source enthusiasts, healthcare workers, students, aid workers, apart from Google Summer of Code interns and aspirants.

Electronic Medical Records, or EMRs, play a vital role in the field of healthcare. The act of moving patient records from traditional paper-based filing systems to a computer-based medical record system has led to a wide range of benefits for physicians and patients alike. These benefits include better quality of healthcare, improved management of patient information and treatment services, as well as more efficient access to patient data for analysis and prediction during research.

There is a perception that EMRs are a luxury only developed nations can afford. This is not true. EMRs play a critical role in improving the quality of care in any country, developed or not. In fact, they are essential for improving treatment facilities in underdeveloped nations. Therefore, EMRs should be considered a mandatory requirement for underdeveloped nations and not merely as a 'non-essential' tool to improve luxury standards.

Developing FOSS software is no easy task. And developing open source healthcare systems is even harder. The problem in developing FOSS healthcare software (such as EMRs) is two-fold. First, EMRs are safety-critical applications, which must adhere to the most stringent quality standards. Second, they require thorough inputs from medical practitioners and healthcare providers who are expected to use the final product. A successful EMR application should provide user friendliness

as well as quality. It should represent the collaborative efforts of both software engineers and healthcare professionals.

However, getting such resources and commitment can be extremely difficult for a start-up. Unfortunately, many open source start-ups are led by moonlighting FOSS enthusiasts, who lack both time and resources to devote to their projects. A majority of these projects will ultimately fail, despite their best efforts. Therefore, it is very rare to see an open source healthcare product such as OpenMRS become a leader in its field, and go on to become the preferred solution for healthcare facilities across the developing world.

The history of OpenMRS: A five-minute guide

OpenMRS was launched in 2004 as an open source medical record system for the developing world. OpenMRS is a global non-profit community with individual an organisational volunteers from around the world. It was founded by two US-based organisations—Regenstrief Institute, a long-time leader in medical informatics research, and Partners In Health, a philanthropic organisation.

The very first OpenMRS implementation was launched in Kenya by the Academic Model Providing Access To Healthcare (AMPATH). This implementation was launched at Eldoret, Kenya in February 2006. PIH introduced OpenMRS to Rwanda in August 2006, while the South